

STAMATIS CHOUDALAKIS

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SUMMARY

I am a mathematician, currently pursuing my PhD in bioinformatics. I have a strong foundation in machine learning algorithms, specializing in clustering analysis and network graph theory. I have proven experience in data curation, feature engineering, and developing robust, reproducible data science and machine learning pipelines in Python and R. Military obligations are fulfilled.

EXPERIENCE

Freelancing

Athens, Greece

Sep 2020 – Present

Provided comprehensive courses in diverse mathematical fields to over 80 undergraduate and postgraduate students. I also guided students through their assignments in Python, R, and VBA, covering aspects of development and data science projects, including data curation, statistical analysis, mathematical modelling, and effectively communicated complex technical subjects in an accessible manner.

Elementary School Teacher

Zygouraki-Leontioy Centre of Creative Activities for Children | Athens, Greece

Jan 2024 – Jun 2024

Tutored and supported 8+ elementary students per hour in completing math assignments.

Ballroom Dance Instructor

The Dance Club, The Mall of Dance | Athens, Greece

Sep 2016 – Jul 2022

Taught, choreographed, and competed in Latin, European, and Modern ballroom dances with students aged 15-65. Mentored 5 students who are now professional dancers.

EDUCATION

PhD in Bioinformatics

Mathematics Research Center, Academy of Athens | School of Medicine, NKUA

Mar 2024 – Present

Thesis: "Cancer driver gene identification using machine learning".

Identified 42 previously unreported mutational patterns in cancer initiation and progression. Built state-of-the-art machine learning and data curation pipelines using Python and R. Optimized computational workflows and solved critical bottlenecks. Created rigorously tested and clean code for reproducible results.

MSc in Applied Mathematics (Grade: 8.11/10)

Department of Mathematics, NKUA

Sep 2020 – Jun 2022

Syllabus included advanced topics in statistical simulation, statistics of machine learning, numerical methods for ODEs, and numerical linear algebra. Dissertation: "Network Graphs of Cancer Mutations" in <https://pergamos.lib.uoa.gr/uoa/dl/object/3237319>.

BSc in Mathematics (Grade: 6.29/10)

Department of Mathematics, NKUA

Sep 2015 – Sep 2020

LECTURES

Lectures on Machine Learning

School of Medicine, NKUA

Mar 2024 – Present

TECHNICAL SKILLS

Programming Languages

Developed machine learning pipelines, including the data preprocessing and visualization in Python (Code is freely accessible upon request: https://github.com/Stchoud/patterns_intra_clustering).

Developed data preprocessing pipelines in R (Code is freely accessible upon request: https://github.com/Stchoud/multiomic_intra_clustering).

Other programming skills include leveraging Bash for routine task automation, foundational knowledge of SQL, and scripting in MATLAB for mathematical modelling.

Machine Learning

Clustering Analysis, Dimensionality Reduction, Classification, Regression, Feature Engineering, Model Evaluation, Cross-validation, Hyperparameter Tuning.

Data Analysis and Visualization

Data Curation (pandas, r-base), Exploratory Data Analysis (matplotlib, ggplot2), Network Visualization (Gephi, Cytoscape), Statistical Modelling

Tools

VS Code, Git, LaTeX, Docker, Jupyter Notebook, SSMS, SSH.

PUBLICATIONS

Choudalakis S., Kastis G. A., & Dikaio N. (2025). Intra-clustering analysis reveals tissue-specific mutational patterns. Computer Methods and Programs in Biomedicine, 263, 108681. (<https://doi.org/10.1016/j.cmpb.2025.108681>)

Choudalakis S., Mitrouli M., Polychronou A., & Roupa P. (2021). Solving high-dimensional problems in statistical modelling: a comparative study. Mathematics, 9(15), 1806. (<https://doi.org/10.3390/math9151806>)

FOREIGN LANGUAGES

English

Level C2, ECPE, University of Michigan

Jun 2013

Italian

Level B1, National Foreign Language Exam System

Jun 2012